

Simple Cryptocurrency using Blockchain

College Project Presentation

Abstract

- This project demonstrates a simple cryptocurrency system to help understand blockchain concepts. It covers wallet creation, transactions, and maintaining a ledger.

Introduction

- Cryptocurrencies are digital currencies built on blockchain technology.
 - - Ensure transparency and security
 - - Use decentralized ledgers
 - - Example: Bitcoin
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- This project implements a basic cryptocurrency to demonstrate these concepts.

Objectives

- Understand blockchain and cryptocurrency basics
- Create a simple cryptocurrency model
- Simulate wallet transactions
- Implement a simple ledger system

Methodology

- 1. Create wallets (simulated user IDs)
- 2. Allow transactions between wallets
- 3. Record transactions in a ledger (block)
- 4. Link blocks together (blockchain)
- 5. Display ledger for transparency

Implementation

- Language: Python
- Modules: hashlib (for hashing), JSON (for ledger)
- Features:
 - Create wallets
 - Add transactions
 - Generate blocks
 - Verify chain integrity

Expected Outcomes

- Users can create wallets and perform transactions
- Transactions are recorded in a transparent ledger
- Better understanding of cryptocurrency working
- Hands-on blockchain fundamentals

Applications

- - Understanding cryptocurrency fundamentals
- - Academic project use
- - Base for advanced blockchain applications

Future Scope

- Add cryptographic keys for security
- Implement mining mechanism
- Extend to NFT or DeFi applications

Conclusion

- The project successfully demonstrates a simple cryptocurrency system.
- It provides insights into blockchain operations and practical learning.
- Can be extended into real-world blockchain applications.